CHAPTER

"Early" Constrained Reconstruction Methods

5

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5.1 Introduction

Constrained image reconstruction has a long history in MRI and an even longer history outside of MRI. For example, there was more than enough material to write a longer than 100-page review article on constrained MRI nearly 30 years ago [79], and the field has continued to see vigorous development over the intervening decades. Although many of the implementation details have evolved over time, the basic concepts and principles from the early literature are still relevant and are frequently embedded in the modern constrained image reconstruction methods. In this chapter, we will discuss some of the key constrained imaging concepts that emerged in the early literature, including those based on support constraints, phase constraints, smoothness constraints, sparsity constraints, linear predictability constraints, and constraints derived from reference scans. To provide notation and context for the discussion, we will briefly review conventional Fourier reconstruction and the relevant literature on early constrained image reconstructions in the remainder of this section.

5.1.1 Basic Fourier reconstruction

For simplicity, our description will focus on 1D Fourier imaging, although all the results extend easily to higher-dimensional imaging scenarios. As discussed in Chapter 2, for an experiment that acquires M k-space measurements at sampling positions k_m for $m = 0, 1, \ldots, M-1$, the measured data $\hat{s}(k_m)$ is related to the original image function $\rho(r)$ by the noisy Fourier transform model:

$$\hat{s}(k_m) = s(k_m) + \eta_m = \int_{-\infty}^{\infty} \rho(r) e^{-i2\pi k_m r} dr + \eta_m, \tag{5.1}$$

where η_m represents measurement noise and $s(k)$ is the ideal noiseless Fourier data.

Without additional assumptions, the image reconstruction problem corresponding to the imaging model from Eq. (5.1) is inherently ill-posed. In particular, there are infinitely many possible images $\rho(r)$ that will be perfectly consistent with the measured data samples $\hat{d}(k_m)$, and when accounting for noise, there is an even larger set of images that will be nearly data-consistent. To obtain a unique reconstruction, it is thus necessary to make additional assumptions about the image $\rho(r)$, which is a cornerstone of *constrained image reconstruction*.

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The classical Fourier reconstruction method [77] is based on the finite sum

$$\hat{\rho}(r) = \mathbb{1}_{FOV}(r) \sum_{m=0}^{M-1} w_m \hat{s}(k_m) e^{i2\pi k_m r}, \tag{5.2}$$

where the weights w_m can all be set to the same value when data is sampled uniformly [77] or can be set to different values to account for nonuniform sampling density [55] (further details can be found in Chapter 4), and $\mathbb{1}_{FOV}(r)$ denotes the indicator function for the field-of-view (FOV). One way to justify Eq. (5.2) is to view it as a simple Riemann sum approximation of the continuous inverse Fourier transform integral, which would have been an analytic perfect-reconstruction formula if we had access to noiseless Fourier data $s(k)$ at all possible (continuous) k-space locations. In addition, assuming that the support of $\rho(r)$ is wholly within the FOV (i.e., $\rho(r) = 0$ for $r \notin [-W/2, W/2]$, where W defines the size of the FOV) and k-space is sampled uniformly at the Nyquist rate (i.e., $\Delta k = 1/W$), then the reconstruction given by Eq. (5.2) with uniform weights corresponds to the unique "minimum-norm reconstruction" [77,131], which has the smallest $\mathcal{L}_2$ norm (i.e., the smallest signal energy) among all data-consistent reconstructions supported within the FOV.

Several notable properties of the Fourier reconstruction can be obtained from the following equation that relates $\hat{\rho}(r)$ obtained from Eq. (5.2) to the true image $\rho(r)$:

$$\hat{\rho}(r) = \int_{-\infty}^{\infty} \rho(\tau) h(r-\tau) d\tau + \sum_{m=0}^{M-1} w_m \eta_m, \tag{5.3}$$

where the point-spread function (PSF) is defined by

$$h(r) \triangleq \sum_{m=0}^{M-1} w_m e^{i2\pi k_m r}. \tag{5.4}$$

- *Resolution*: With uniform weights and uniform Nyquist-rate sampling with symmetric k-space coverage, it can be shown that the PSF for Eq. (5.2) satisfies $h(r) \propto \frac{\sin(\pi M \Delta k r)}{\sin(\pi \Delta k r)}$. Convolution with $h(r)$ leads to spatial blurring. While there are multiple ways to quantify spatial resolution, a common approach based on the effective width of $h(r)$ results in a nominal spatial resolution of $1/(M\Delta k)$ for this case. *Reconstructing images with spatial resolution that is better than this is a "super-resolution" reconstruction problem.*
- *Undersampling*: When k-space sampling violates the Nyquist sampling criterion, i.e., $\Delta k > 1/W$, the PSF will have undesirable characteristics that lead to various kinds of aliasing artifacts. *Reconstructing aliasing-free images from sub-Nyquist data is a "sparse data" reconstruction problem.*
- *Noise*: Let η_m be independent identically distributed noise with zero mean and standard deviation σ. With uniform weights in Eq. (5.2), the image-domain noise standard deviation will be the same at every spatial location and proportional to $\sqrt{M}\sigma$. Thus, if one attempts to improve image resolution by extending k-space coverage from $M\Delta k$ to $2M\Delta k$ by doubling the data acquisition time (already a major sacrifice!), the image signal-to-noise ratio (SNR) will also be $\sqrt{2}\times$ worse. *Reconstructing high-quality images from noisy data is a "denoising" reconstruction problem.*

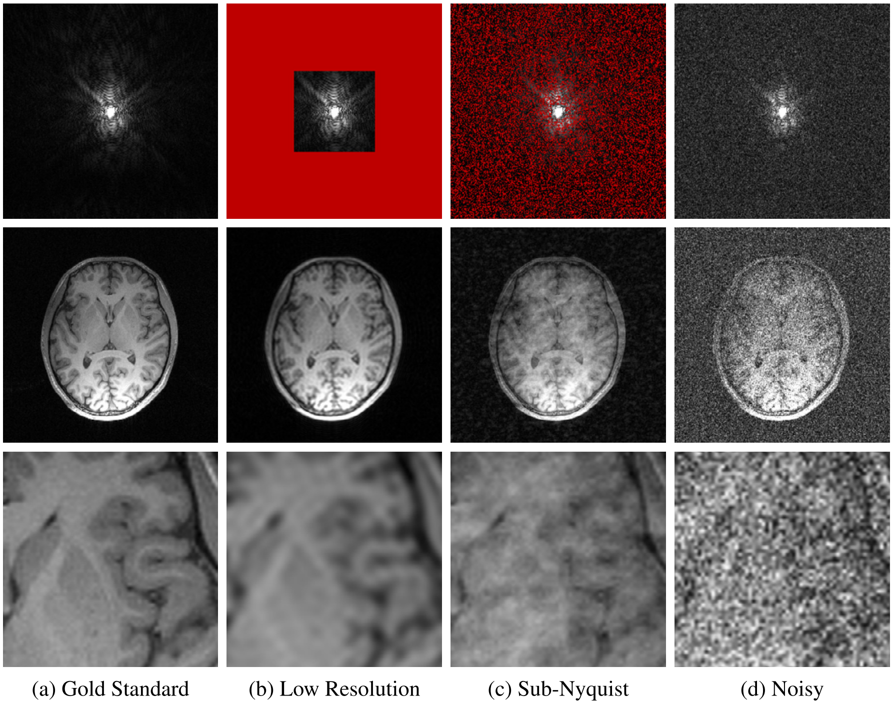


FIGURE 5.1

Illustration of the degradations in image quality for traditional Fourier reconstruction corresponding to (b) low-resolution acquisition, (c) sub-Nyquist acquisition, and (d) noisy data. The top row shows k-space data (with missing samples indicated in red (gray in print version)), while the middle row shows the corresponding reconstructed MRI images, and the bottom row shows image zoom-ins for enhanced visualization.

Illustrations of typical image quality degradations resulting from resolution, undersampling, and noise are shown in Fig. 5.1. The early literature on constrained image reconstruction proposed various techniques to mitigate these degradations, as will be reviewed in the next subsection.

5.1.2 Constrained reconstruction: historical perspective

Constrained image reconstruction has a long history and wide applications that go well beyond MRI. This subsection provides a brief literature review. Because of the large amount of literature that exists, this review is by no means comprehensive. The goal is to provide context for discussing the methods described in Sections 5.2–5.7.

For many years before the invention of MRI, it had been believed that the resolution of a Fourier imaging system was fundamentally limited by k-space coverage. This led to concepts like the Rayleigh limit to spatial resolution, which was based on the characteristics of the PSF [66,123]. However, it later became recognized that missing or degraded frequency information could actually be recovered through the process of constrained reconstruction by utilizing additional prior information [66,103,123].

The concept of constrained reconstruction was used extensively in other modalities before it was first applied to MRI. Some of these early approaches were focused on achieving superresolution by utilizing detailed prior knowledge about the exact spatial support of the image within the FOV, together with the fact that support-limited images will have analytic k-space data (such that the Fourier signal must be smooth). Slepian, Landau, and Pollak's extensive work on support-limited functions and prolate spheroidal wave functions [69,70,115–117] was of significant theoretical value, and the Gerchberg–Papoulis algorithm was perhaps the first practical method to perform support-limited extrapolation for both continuous and discrete data in the presence of measurement noise [26,99]. Other approaches were based on the idea that, when presented with ambiguity and uncertainty about the true solution to an inverse problem, we should avoid choosing solutions that appear to be more complicated or definitive than might be supported by the measured data, and should instead have a preference for solutions that are as simple and noncommittal as possible. This is a general philosophical principle that is sometimes known as Occam's razor. Although measuring the "simplicity" of a reconstruction is subjective, some of the earliest common approaches include a preference towards solutions that have minimal signal energy or maximal spatial smoothness [3,8,24,25,47,122], maximum entropy [31,57], sparsity [3,8,9,24,25,47,48], or nonnegativity [83,108].

In the early days of MRI, constrained reconstruction research was mainly focused on solving the super-resolution problem. Some notable constraints that were used for this purpose include support constraints [79,105,131], phase constraints [17,79,89,94], linear predictability constraints [32,38,78, 79,118], and constraints derived from reference images [12,52,60,74,76,132]. However, it was later recognized (particularly after the emergence of parallel imaging and compressed sensing) that many of these constraints used for superresolution reconstruction could be used much more effectively when applied to sub-Nyquist reconstruction [12,19,30,86,90,106,107] or denoising reconstruction [33,37, 41] problems. As a result, recent research in constrained image reconstruction has shifted away from addressing the superresolution problem per se.

Although sub-Nyquist reconstruction rose to popularity more recently within MRI, there were substantial early theoretical foundations for the recovery of general signals from sub-Nyquist measurements, including Landau's interpolation theory [68] and Papoulis' multichannel sampling theory [100]. In the signal processing community, much work was done by Bresler's group on the development of practical sampling schemes and reconstruction algorithms to achieve the Landau bound [133].

For static MRI applications, Cao and Levin's early Feature-Recognizing MRI approach from 1993 [12] not only utilized sub-Nyquist sampling, but also represents an important precursor to modern data-driven machine learning constrained MRI reconstruction methods [64,81,111]. Feature-Recognizing MRI took a database of training images of the same body part from multiple subjects, and used standard low-rank modeling/principal component analysis [59] to identify a low-dimensional set of basis functions that optimally captured most of the energy of these images. This low-dimensional representation could then be used to reconstruct unseen data of the same body part from a new subject, and could also be used to derive optimal k-space sampling locations (that turned out to be nonuniform). Reeves [107] observed that uniform Nyquist sampling was suboptimal for support-constrained reconstruction and optimized a set of irregularly-spaced k-space sampling locations using estimation-theoretic (Cramér–Rao bound) concepts. Similarly, von Kienlin and Mejia [134] optimized k-space sampling locations under the modeling assumptions of the earlier reference-based SLIM constrained imaging approach [52]. Marseille et al. [90] used Cramér–Rao bounds to discover that, if images were modeled as smooth regions separated by edge structures occurring at unknown locations, then k-space interpolation (i.e., reconstruction of sub-Nyquist data) would be theoretically more reliable than k-space extrapolation (i.e., the super-resolution problem). This can be seen to be conceptually quite similar to more recent compressed sensing approaches that impose sparsity constraints on image edges [7,42,86,124]. Dologlou et al. [19] later combined this nonuniform sampling concept with structured low-rank Hankel matrix concepts, resulting in what was possibly the first linear-predictive structured low-rank matrix completion method for sub-Nyquist constrained MRI reconstruction and appeared long before the recent reinvigoration of interest in this kind of approach [5,34,38,39,58,96,113].

For dynamic MRI applications, Xiang and Henkelman introduced the (k,t)-space formulation in 1993 [137]. Based on the (k,t)-space formulation, it was shown by Liang and Lauterbur in 1997 that motion artifacts were due to temporal undersampling and high-quality image reconstruction from undersampled (k, t)-space data was possible using special temporal basis functions [80]. Many methods were developed for accelerated dynamic imaging using sparse sampling of (k, t)-space: the UNFOLD method [87,128], the (k,t)-BLAST method [129], and the partial separability model-based method [14,15,36,73,138] are just three examples of many methods that were developed to provide practical means to undersample (k,t)-space and to reconstruct high-quality images from sub-Nyquist (k,t)-space data. Some of these methods also serve as the basis for the more recent dynamic low-rank (Casorati) matrix and tensor recovery methods [36,45,82,97,126,127,138]. There were also a number of early methods that accelerated dynamic imaging using sub-Nyquist sampling without invoking the (k,t)-space formulation. One example is the use of spatial-support constraints to achieve sub-Nyquist sampling [49,93], while others include the TRICKS and HYPR methods developed by Mistretta's group, which have been widely used for high-speed angiographic imaging [91,92].

Compared to super-resolution and sub-Nyquist reconstruction, there was relatively little work on the denoising reconstruction problem in the early days of MRI, even though limited sensitivity and low SNR have long been major limitations of MRI signal detection [77]. Although there was a substantial amount of early work on the denoising reconstruction problem for other imaging modalities [8,13,23,25,28,47,71,119], the early MRI work generally treated reconstruction and denoising as two distinct steps instead of combining them together. As a result, the early MRI denoising work often relied on standard image processing tools like wavelet denoising [46,95,104,135,136] and edge-preserving smoothing using anisotropic diffusion methods [27,109,114] or Markov random fields [53], low-rank matrix modeling/principal component analysis methods for denoising time series data [121], or special data acquisition schemes to improve SNR [10,20,88,101,102]. More recently, it has been shown that it is beneficial to acquire noisy high-frequency k-space data and perform denoising and image reconstruction jointly [33,37,41,43]. Such an approach has proved effective for a number of SNR-limited imaging applications, such as high-resolution MR spectroscopic imaging [67] and diffusion imaging [44].

5.2 Support-constrained reconstruction

The support of an image is the set of spatial locations r for which $\rho(r) \neq 0$. Most MRI reconstruction methods employ a weak form of support constraint, assuming that $\rho(r) = 0$ outside of the FOV. However, when we refer to "support-constrained" methods, we specifically mean methods that make the stronger assumption that the image $\rho(r)$ does not actually occupy the entire FOV. Instead, it is assumed that there are regions within the FOV for which $\rho(r) \approx 0$. Mathematically, this can be described as

$$\rho(r) \approx 0 \text{ for } r \notin \Omega, \tag{5.5}$$

where the set Ω denotes the support of the image and is assumed to satisfy $\Omega \subset [-W/2, W/2]$.

In the early literature, it was particularly common for the image support to be known *a priori* from reference images. Given this knowledge, constrained reconstruction could be performed using, e.g., linear algebraic formulations that only allowed voxels within the support to be nonzero or iterative algorithms that alternatingly projected the current image estimate onto (i) the set of data consistent images and (ii) the set of support-limited images. This latter approach can be viewed as a form of the Projection-Onto-Convex Sets (POCS) algorithm [79], which can have nice theoretical convergence characteristics [2].

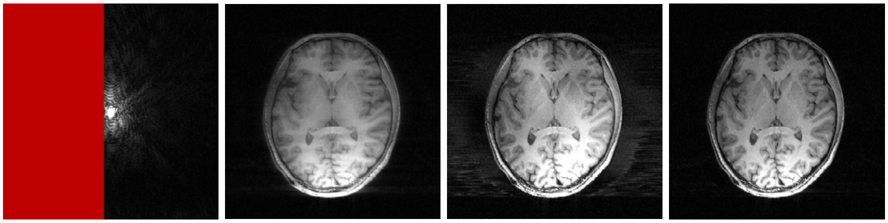
Although the details are somewhat technical, these approaches can be understood at a high level because of the Fourier transform property that support-limited images $\rho(r)$ will always have Fourier transforms $s(k)$ that are analytic (i.e., smooth and infinitely differentiable). This smoothness characteristic implies that it can be possible to extrapolate or interpolate unmeasured k-space data from incomplete or noisy k-space data.

While support constraints have been applied to extrapolate missing high-frequency information in MRI scenarios [79,105,131], this super-resolution approach is limited by the fact that it tends to be ill-posed and very sensitive to noise, leading to practical gains in image spatial resolution that are not too significant. However, it should be noted that support constraints are still widely used in modern constrained MRI methods, although, for sub-Nyquist reconstruction applications rather than the super-resolution scenarios they were originally developed for. In some cases, it can be hard to see that support constraints are being used because they are implicit rather than explicit. For example, in multichannel parallel imaging methods (see Chapter 6 for more information) like SENSE [106], image support constraints are often an implicitly incorporated into the definition of coil sensitivity maps [130]. It has also been recently shown [34,38] that virtually all linear predictive methods (including methods like GRAPPA [30], SPIRiT [84], and structured low-rank approaches [5,34,38,39,58,96,113]) will implicitly take advantage of image support constraints when reconstructing support-limited images.

5.3 Phase-constrained reconstruction

Phase-constrained reconstruction was one of the earliest successful approaches to constrained reconstruction in MRI. These methods were originally developed to solve the data extrapolation problem associated with half-Fourier (or partial-Fourier) imaging. As the "half-Fourier" name implies, here $s(k)$ is measured only for half k-space, say, $\forall k \geq 0$.

To understand how phase-constrained extrapolation works, consider the simple scenario in which the image function $\rho(r)$ is purely real-valued, with zero imaginary component. In this case, $s(k)$ has perfect Hermitian symmetry, i.e., $s(-k) = s^*(k)$, where * denotes complex conjugation. This symmetry relationship suggests a simple data extrapolation scheme to generate $s(-k)$. Indeed, this approach corresponds to the earliest form of half-Fourier imaging [22]. In practice, $\rho(r)$ is not real-valued due to field inhomogeneity, coil sensitivity variations, flow effects, etc., and more practical data extrapolation methods were developed to cope with the problem, using a pre-estimated phase $\hat{\phi}(r)$ as a constraint.



(a) Partial Fourier (b) Fourier (c) Margosian (d) POCS

FIGURE 5.2

Illustration of single-channel phase-constrained partial Fourier reconstruction using the Margosian and POCS algorithms. In this case, (a) a half-Fourier sampling scheme was augmented with an additional set of eight central k-space lines to enable phase estimation. While (b) classical zero-filled Fourier reconstruction of this data is heavily blurred, the (c) Margosian and (d) POCS reconstructions of this data are quite successful at recovering crisp high-resolution image features, although the Margosian method especially demonstrates visible artifacts stemming largely from imperfect phase estimation.

For example, the imaging model from Eq. (5.1) could be replaced with

$$\hat{s}(k_m) = \int_{-\infty}^{\infty} m(r) e^{i\hat{\phi}(r)} e^{-i2\pi k_m r} dr + \eta_m \tag{5.6}$$

where $m(r)$ represents the unknown magnitude image. The fact that $m(r)$ is real-valued means that it has roughly half the degrees of freedom as the complex-valued image $\rho(r)$, resulting in a substantially easier inverse problem.

Assuming that the phase is relatively smooth (which is a good assumption in many cases, though is not always appropriate), such a phase constraint could, e.g., be obtained by collecting a few additional samples across the center of k-space to provide a symmetric data set to estimate $\hat{\phi}(r)$. Imposing this phase constraint along with the data-consistency constraint leads to one version of the phase constrained reconstruction problem. Several methods were proposed for solving this problem including the Margosian method [89], the homodyne detection method by Noll and Macovski [94], the Cuppen method [17], and the POCS method [79]. We have already mentioned the POCS algorithm in the context of support-constrained reconstruction. While the POCS algorithm for phase-constrained reconstruction is different, it still follows the basic principle of iteratively projecting the current estimate of the image onto two different constraint sets. In the context of phase-constrained reconstruction, the image is iteratively projected onto (i) the set of data consistent images and (ii) the set of images that are consistent with the estimated phase $\hat{\phi}(r)$. An illustration of the performance of the Margosian and POCS methods is shown in Fig. 5.2.

Although the phase-constrained extrapolation problem was actively addressed 30+ years ago, the basic concepts of using prior phase information and leveraging Fourier symmetry for constrained reconstruction are still relevant today. While some of the more recent approaches bear relatively close resemblance to some of the earliest methods by relying explicitly on prior phase estimates [11,110], recent years have also seen the introduction of very different kinds of phase constraints that have very different structure. For example, recent years have witnessed the emergence of phase-constrained reconstruction methods that do not actually require any explicit phase-estimation procedure, which eliminates one of the major pitfalls for classical methods. Instead, these newer methods are based on concepts of linear-predictability across different sides of k-space [34,38,54], virtual conjugate coils [6], and structured low-rank matrix recovery [34,38,39,62]. Compared to earlier approaches, these new types of approaches have substantially more flexibility than previous approaches and, for example, are capable of exploiting Fourier symmetry relationships with very different sampling patterns than before. This flexibility allows them to not only be used for super-resolution reconstruction of Nyquist-sampled data (as was the case for the classical methods), but also allows them to be used for reconstruction of data sampled below the Nyquist rate with samples missing from both sides of k-space and without dense sampling of the k-space center [34,38,39,62].

5.4 Linear predictive reconstruction

The autoregressive moving average (ARMA) model was one of the first methods that used linear prediction to extrapolate k-space data [118]. In MRI, a (P, Q)-order ARMA model of an image $\rho(r)$ will often take the form

$$\rho(r) = \frac{\sum_{q=0}^{Q} \beta_q e^{i2\pi \Delta k q r}}{1 - \sum_{p=1}^{P} \alpha_p e^{i2\pi \Delta k p r}}. \tag{5.7}$$

In k-space, this is equivalent to

$$s(n\Delta k) = \sum_{p=1}^{P} \alpha_p s([n-p]\Delta k) + \frac{1}{\Delta k} \sum_{q=0}^{Q} \beta_q \delta[n-q], \tag{5.8}$$

where $\delta[n]$ is the Kronecker delta function. One may notice that Eq. (5.8) is a generalized version of linear prediction that allows one sample in k-space to be predicted from previous neighboring values, which is a perfect fit for extrapolating higher-frequency k-space data from a set of Nyquist-sampled low-frequency k-space measurements. It should also be noted that the ARMA model from Eq. (5.7) reduces to a conventional Fourier model in the form of Eq. (5.2) when $P = 0$ and to a strict autoregressive model when $Q = 0$. Combining both parts of this model together enables the representation of both smooth features (via the coefficients β_q) and spiky/support-limited features (which autoregressive models are excellent at capturing [38]).

There were a number of extensions on the ARMA model for linear predictive extrapolation. One notable approach, proposed by Liang and Haacke [32,78], applies the ARMA model to a weighted version of the k-space data, i.e., $\tilde{s}(k) = (i2\pi k)^p s(k)$ with $p \geq 1$. In the image domain, this weighting of k-space is equivalent to applying a spatial differentiation operator, which is a common sparsifying transform in the modern literature. It was shown that, if the image function is piecewise constant or piecewise polynomial, the ARMA model is precise after such a sparsifying transform; in other words, the k-space data is perfectly linear predictable after this sparsifying transform [72,78]. Interestingly, this work may be viewed as some of the earliest attempts to use sparsity to constrain image reconstruction in MRI and also represents some of the first applications of structured low-rank Hankel matrix modeling in this field [78] (see also Chapters 8 and 10).

All of the early linear predictive models were originally formulated for 1D extrapolation of k-space in order to achieve super-resolution. Unfortunately, while these types of approaches were reasonably effective with noiseless data, 1D extrapolation over long distances can be quite challenging and very sensitive to small amounts of noise. Modern linear predictive methods can not only handle multidimensonal k-space data but also sub-Nyquist data [38], significantly enhancing its practical utility. For example, the popular multichannel parallel imaging (see also Chapter 6) methods GRAPPA [30] and SPIRiT [84] are based on multidimensional linear predictive relationships, and the constrained imaging field is currently enjoying somewhat of a renaissance of linear predictive approaches [38] due to the reemergence of structured low-rank matrix recovery approaches. These latter approaches are quite powerful in that they enable the automatic data-adaptive identification and imposition of a large number of simultaneous imaging constraints, including support constraints, phase constraints, sparsity constraints, parallel imaging constraints, and multicontrast constraints, often without requiring any kind of calibration data [5,34,38,39,58,96,113].

5.5 Rank-constrained reconstruction

MR signals often possess low-rank structures when put in the form of Hankel matrices or Casorati matrices (or tensors) due to linear predictability [38] and partial separability [36,73,112]. These low-rank structures have been exploited in a number of early constrained reconstruction methods for addressing all three of the main constrained reconstruction problems: super-resolution reconstruction, sub-Nyquist data reconstruction, and denoising. This section will describe these low-rank structures and their use for constrained image reconstruction (see also Chapters 9 and 10).

Consider a sequence $s[n]$ of uniformly-spaced k-space samples such that $s[n] \triangleq s(n\Delta k)$, and assume that this sequence satisfies the following shift-invariant autoregressive relationship

$$s[n] = \sum_{p=1}^{P} \alpha_p s[n-p] \tag{5.9}$$

for all possible integer values of n. Note that this corresponds to a form of linear predictive model as described in the previous section. In addition, either general support constraints [34,38] or a specific kind of sparsity constraint [72,78] can be used to guarantee the accuracy of this autoregressive model. If a Hankel or Toeplitz matrix is formed from this kind of data, it can be shown that the matrix will have low-rank characteristics. For example, for any chosen integers N and L with $N > L$, the Hankel matrix $\mathbf{H} \in \mathbb{C}^{(N-L)\times L}$ constructed as

$$\mathbf{H} \triangleq \begin{bmatrix} s[0] & s[1] & s[2] & \dots & s[L-1] \\ s[1] & s[2] & s[3] & \dots & s[L] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s[N-L-1] & s[N-L] & s[N-L+1] & \dots & s[N] \end{bmatrix} \tag{5.10}$$

must have $\mathrm{rank}(\mathbf{H}) \le \min\{P, L, N-L\}$. If N and L are chosen such that $L \gg P$ and $N-L \gg P$, then $\mathbf{H}$ will be heavily rank-deficient.

High-dimensional MR signals are also often partially separable. For example, in dynamic imaging, the ideal k-t space data $s(k,t)$ can often be modeled as being Qth-order partially separable:

$$s(k,t) = \sum_{q=1}^{Q} g_q(k) h_q(t). \tag{5.11}$$

If a Casorati matrix is formed from this kind of data, it can be shown that the matrix will also have low-rank characteristics. In this case, for any chosen integers M and N, the Casorati matrix $\mathbf{C} \in \mathbb{C}^{M\times N}$ constructed as

$$\mathbf{C} \triangleq \begin{bmatrix} s(k_1,t_1) & s(k_1,t_2) & \dots & s(k_1,t_N) \\ s(k_2,t_1) & s(k_2,t_2) & \dots & s(k_2,t_N) \\ \vdots & \vdots & \ddots & \vdots \\ s(k_M,t_1) & s(k_M,t_2) & \dots & s(k_M,t_N) \end{bmatrix} \tag{5.12}$$

must have $\mathrm{rank}(\mathbf{C}) \le \min\{Q, M, N\}$. If M and N are chosen such that $M \gg Q$ and $N \gg Q$, then $\mathbf{C}$ will be heavily rank-deficient.

The rank constraints can be enforced in a number of ways in constrained reconstruction, both in the early literature and in modern efforts. One approach that has become increasingly common in the modern literature is the use of regularization penalties that prefer lower-rank matrices, although there remain many viable alternatives that may be more or less useful depending on the specific application. Although the concepts of low-rank matrix modeling have been used in MRI for decades, recent years have seen a particular resurgence of interest in structured low-rank matrix recovery methods that use the characteristics of Hankel/Toeplitz matrices to enforce support, phase, sparsity, parallel imaging, and multicontrast constraints [5,34,38,39,58,96,113] without requiring the kinds of calibration data that would often be employed by early methods when using such constraints. In addition, methods based on the low-rank structure of Casorati matrices/tensors [36,45,82,97,126,127,138] are also quite common in the modern literature, particularly in dynamic and multicontrast imaging contexts. Similar to the case of Hankel/Toeplitz low-rank modeling, one of the important differences between the modern literature and the early literature is that modern techniques are frequently capable of identifying the low-rank structure directly from sparsely sampled or low-quality measurements, without requiring reference information or calibration data to pre-estimate the low-rank structure.

5.6 Sparsity-constrained reconstruction

Sparsity-constrained reconstruction is very visible within the modern constrained MRI reconstruction literature due to the rise of compressed sensing concepts more than a decade ago [7,42,61,85,86,124]. However, there had already been substantial work in sparse modeling for constrained MRI before that time. While modern compressed sensing approaches are frequently defined by their use of sparsity together with nonlinear reconstruction and some form of quasi-random/incoherent sub-Nyquist sampling scheme, the early MRI methods frequently did not share these same features, oftentimes relying on sampling patterns that were more uniform and reconstruction methods that were much more linear than modern approaches.

In order to define a simple notion of sparsity, we will assume that the set of functions $\{\phi_n(r)\}$ forms a basis or frame for the space of possible images. In other words, for any possible image $\rho(r)$, there exists a sequence of coefficients $\{\alpha_n\}$ such that

$$\rho(r) = \sum_n \alpha_n \phi_n(r). \tag{5.13}$$

An image $\rho(r)$ is said to have a sparse representation in the domain specified by $\{\phi_n(r)\}$ if most of the α_n coefficients in this representation are equal to zero, with only a small number of nonzero values.

Different approaches were used in the early literature to impose sparsity constraints. One common approach was to pre-estimate the index values where the coefficients α_n would take on nonzero values. This could be done using the low-rank characteristics of Hankel matrices for certain types of sparse signals as described previously [72,78,80], or by using reference images to pre-identify the basis functions that would capture the important structural characteristics of the image [12,52,98]. Once the small number of relevant basis functions has been identified, it is straightforward to use simple linear algebra to estimate the corresponding α_n values. This can be viewed as a form of support-limited reconstruction, where the support constraints are defined with respect to the representation coefficients α_n, rather than original image $\rho(r)$. Similar ideas used in these early methods can be found in modern multicontrast reconstruction methods that utilize the fact that different images of the same anatomy oftentimes share similar sparsity characteristics [4,18,21,29,35,41,43,56,63,65,125]. Similar information-sharing methods were also developed in the early literature for other imaging modalities [13,23,28,71,119].

Another approach for imposing sparsity was based on the use of edge-preserving Markov random fields or edge-preserving anisotropic diffusion filtering schemes that were intentionally designed to preserve sparse image features such as edges. Although these approaches were utilized to some extent in the early MRI literature [50], they were much more commonly found in the early literature for other imaging modalities [3,8,24,25,47]. Widespread modern reconstruction methods like compressed sensing that rely on sparsity-promoting regularization of image edges using penalties like the convex ℓ_1-norm or the nonconvex ℓ_p-norm with $p < 1$ can all be viewed as Markov random field methods. Of course, as said before, a major difference between the early Markov random field methods and modern compressed sensing is the difference in sampling styles—the early work largely used Markov random field models to address super-resolution and/or denoising problems, rather than the sub-Nyquist reconstruction problems that are the primary emphasis of modern compressed sensing methods (which, as described previously, often leads to a much easier inverse problem, yielding higher-quality reconstruction results).

5.7 Reconstruction using side information

In many imaging applications, such as dynamic imaging and spectroscopic imaging, side information, in the form of reference images, is often available. Keyhole imaging [60,132] represents an early attempt in MRI to use reference data for super-resolution. To illustrate how this is done, let's consider the basic keyhole method as applied in dynamic imaging in which a high-resolution reference data set $s_\mathrm{r}(k)$ is obtained for $k = -N\Delta k, \cdots, N\Delta k$, followed by a sequence of dynamic data sets $s_\mathrm{t}(k)$ for $k = -M\Delta k, \cdots, M\Delta k$, where $M \ll N$. A straightforward method to perform data extrapolation on $s_\mathrm{t}(k)$ is to replace the unmeasured high-frequency data with those from $s_\mathrm{r}(k)$. In other words, we obtain extrapolated data as follows:

$$\hat{s}_\mathrm{t}(k) = \begin{cases} s_\mathrm{t}(k), & \text{for } k \in \{-M\Delta k, \cdots, M\Delta k\} \\ s_\mathrm{r}(k), & \text{otherwise.} \end{cases} \tag{5.14}$$

This method can perform perfect or high-quality data extrapolation if $s_\mathrm{r}(k)$ is equal to or approximately equal to $s_\mathrm{t}(k)$ for $|k| > M\Delta k$. This is the case in some dynamic contrast-enhanced imaging experiments where dynamic signal changes take place mainly in the low-frequency k-space data. However, when this assumption is not valid, this simple keyhole data extrapolation scheme would not be effective. This problem can be addressed using a more sophisticated data extrapolation method that uses $s_\mathrm{r}(k)$ to construct an optimal set of basis functions for $s_\mathrm{t}(k)$ [74,76].

As an example, the generalized-series approach [74,76] represents $\rho_\mathrm{t}(r)$ using a series of basis functions $\{\phi_\ell(r)\}$ as

$$\rho_\mathrm{t}(r) = \sum_{\ell=1}^{L} \alpha_{\mathrm{t},\ell}\phi_\ell(r). \tag{5.15}$$

An optimal set of the basis functions $\phi_\ell(r)$ according to the minimum cross-entropy model selection principle is, to a first-order approximation, given by [74,76]

$$\phi_\ell(r) = q(r)e^{i2\pi\ell\Delta kr}, \tag{5.16}$$

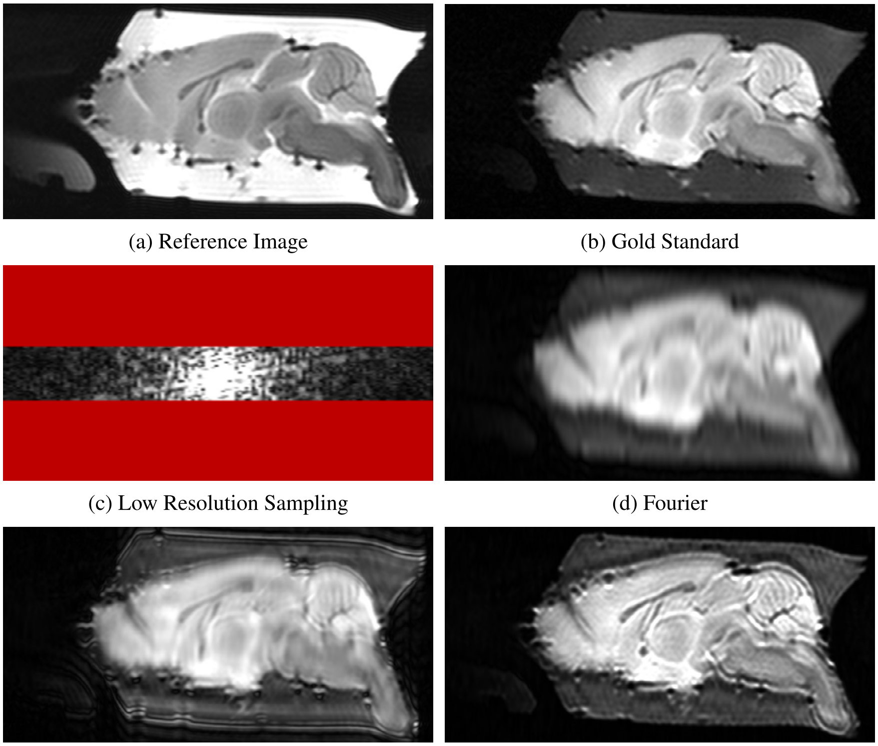
where $q(r)$ can be chosen to equal the reference image $\rho_\mathrm{r}(r)$ or some appropriate constraint function derived from $\rho_\mathrm{r}(r)$.

With the basis functions in Eq. (5.16) and taking $q(r) = \rho_r(r)$, Eq. (5.15) can be rewritten as

$$s_\mathrm{t}(k) = \sum_{\ell=1}^{L} \alpha_{\mathrm{t},\ell} s_\mathrm{r}(k - \ell\Delta k). \tag{5.17}$$

If L is smaller than the number of k-space measurements, the coefficients $\alpha_{\mathrm{t},\ell}$ can easily be estimated using linear algebraic methods (similar to the case described for sparsity-constrained reconstruction with known support in the previous section) from the measured data $s_\mathrm{t}(k)$. After the $\alpha_{\mathrm{t},\ell}$ are determined, Eqs. (5.15) and (5.17) can be used to synthesize the high-resolution dynamic image and/or the corresponding extrapolated dynamic k-space data. A quick illustration of the keyhole and generalized series approaches is given for multicontrast MRI data in Fig. 5.3.

High-resolution reference images have also been used in the early constrained MRI literature to construct other specialized image models. For example, a common situation in MR spectroscopic imaging is that a user may desire to measure accurate information about one specific region of interest (ROI) within the FOV, but the ability to make such a measurement is confounded because the data was acquired with low spatial resolution and there is a substantial amount of signal from undesired regions that leaks into the ROI because of PSF blurring effects for conventional Fourier reconstruction. The Spectral Localization by IMaging (SLIM) [52] technique, which was originally formulated for MR



(a) Reference Image (b) Gold Standard

(c) Low Resolution Sampling (d) Fourier

(e) Keyhole (f) Generalized Series

FIGURE 5.3

Illustration of reference-based image reconstruction using multicontrast data of an ex vivo mouse brain. Given a (a) high-resolution reference image, the goal is to reconstruct (b) an image acquired with different contrast parameters from (c) low-resolution k-space data. (d) Standard Fourier reconstruction has the expected blurring artifacts. (e) Keyhole reconstruction has some potential resolution recovery, though also has substantial artifacts and blurring due to a mismatch between the high-frequency k-space signal between the reference image and the target image. On the other hand, (f) generalized series reconstruction is more tolerant of data mismatches, resulting in artifacts that are less severe. Note that, due to phase mismatches between the reference image and the target image, a low-resolution phase estimate from the target image was used to construct a phase-matched reference image that was used for both keyhole and generalized series reconstruction.

spectroscopic imaging but can also be applied more broadly, is perhaps the earliest attempt to overcome this problem by using information from a high-resolution reference image. In particular, SLIM uses the high-resolution reference image to define a small set of generalized voxels (or compartments) defined by basis functions $\phi_\ell(r)$, where each compartment is designed to exactly match (with high resolution) the edge shapes, sizes, and locations of various image features and regions-of-interest within of the image. The image is then modeled simply by

$$\rho(r) = \sum_{\ell=1}^{L} \alpha_\ell \phi_\ell(r), \tag{5.18}$$

and if the number of voxels is small with the respect to the number of measured k-space samples, the coefficients α_ℓ can be estimated using simple linear algebraic methods. Importantly, rather than directly estimating a high-resolution image, this approach can be interpreted as instead measuring the average value of the image over the spatial region spanned by each compartment [51,75]. Further extensions of this general approach were also proposed in the early constrained MRI reconstruction literature [16,50,74,120,134], and the idea of using prior-edge information derived from high-quality reference images still lives on in more recent approaches that impose constraints in a softer way using regularization [18,21,41,56,65] or that directly attempt to estimate the average signal from a prescribed image compartment using less-stringent modeling assumptions [1,40].

5.8 Discussion

We have reviewed several constrained reconstruction methods, each utilizing one of the key constraints that appeared in the early literature, which include support constraints, phase constraints, linear predictability constraints, rank constraints, sparsity constraints, and constraints derived from reference images. Although more recent methods oftentimes rely on similar constraints to those used in early methods, there are a few distinctions. For one, computational resources were substantially more limited in the early days than they are now. As a result, the early methods tended to focus on simpler and less computationally demanding approaches than the methods that are common today. For example, it was quite common for early methods to focus on 1D reconstruction (e.g., in a 2D acquisition, reconstructing the fully-sampled frequency-encoding dimension using standard Fourier methods, leaving only the phase-encoding dimension to be reconstructed using constraints). Modern approaches that use multidimensional formulations with multidimensional constraints tend to be much more effective than the earlier 1D approaches. Another major difference is that early constrained imaging methods tended to use prior information that was much stronger and more explicit (and subsequently, more prone to failure) than modern methods. For example, early methods that used support constraints, phase constraints, or sparsity constraints often required precise knowledge of the spatial image support, the image phase, and the support of the image in the sparsity domain, which was often obtained from auxiliary reference scans of the subject. This level of strong prior information often enabled the use of computationally inexpensive non-iterative reconstruction methods. In contrast, the use of modern nonlinear/iterative problem formulations allows users to know only that the image possesses some form of limited support, smooth phase, or sparsity in a transform domain —the user no longer needs auxiliary scans to obtain appropriate constraints, since modern approaches are powerful enough to actually extract that kind of information directly from the low-resolution, sub-Nyquist, and/or noisy k-space data. This enables the use of less data and more robustness to inaccurate constraints.

It is also worth noting that, in many MRI applications, one often has flexibility to implement various data acquisition schemes while satisfying the experimental constraints. Different acquisition schemes may lead to a super-resolution problem or a denoising problem or sparse data-recovery problem or their combination. In the old days, the super-resolution problem was the most heavily studied, as researchers often designed data acquisition to sample at the Nyquist rate while protecting SNR. However, from

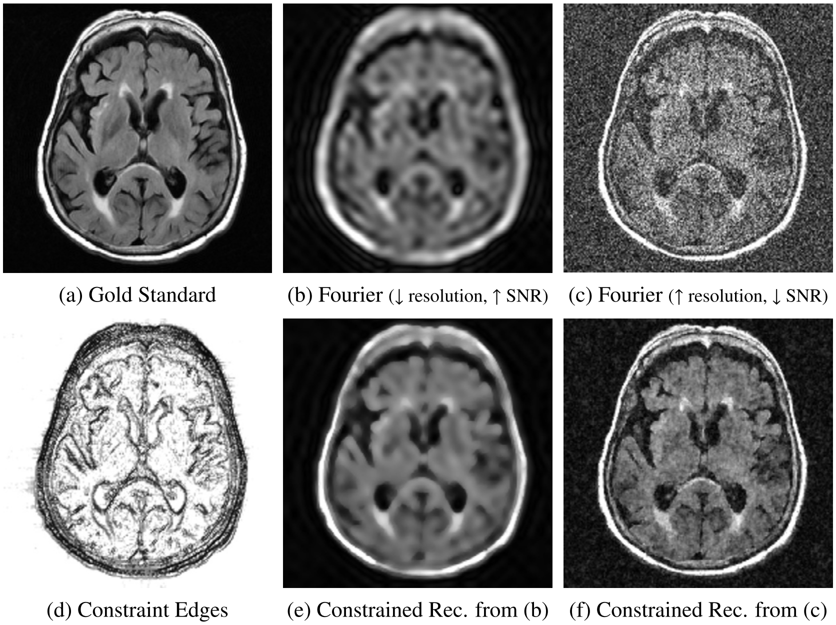


FIGURE 5.4

Illustration of the difference in effectiveness when using the same constraints for different data-acquisition schemes. (a) Gold standard image. (b) Standard Fourier reconstruction of low-resolution high-SNR data. (c) Standard Fourier reconstruction of high-resolution low-SNR data. (d) Image edges (derived from a reference image) that will be used as constraints using the method from Ref. [41]. (e) Constrained reconstruction of low-resolution high-SNR data. (f) Constrained reconstruction of high-resolution low-SNR data.

an image reconstruction standpoint, some of the reconstruction constraints are more effective when combined with some data-acquisition strategies than they are when combined with others. For example, Fig. 5.4 shows a set of results to illustrate the difference in effectiveness when utilizing anatomical boundary constraints (derived from a reference image) for solving a denoising problem (where the data has high spatial resolution but low SNR) compared to solving a super-resolution problem (where the data has low spatial resolution and the extra time was used for averaging to improve SNR). As can be seen, the denoising reconstruction has much better spatial resolution than the super-resolution result. Thus, an important observation from the more recent literature is that, in order to make maximal use of constrained reconstruction ideas, one should formulate and solve the data acquisition problem and the resulting reconstruction problem jointly.

5.9 Summary

Image reconstruction from limited and noisy data is a classical problem that arises in MRI and other imaging applications. A number of constrained reconstruction methods have been developed over the decades to utilize prior information to compensate for the lack of sufficient experimental data and to remove measurement noise. These methods have helped improve the accuracy and efficiency of many MRI techniques and laid a solid technical and conceptual foundation for the development and application of modern image reconstruction methods to be described in later chapters.

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